

Program: Diploma in Food Processing Technology	
Course Code: 5423B	Course Title: Spices and Condiments
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4(L: 4T:0 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand the basics of spices and condiments
2. To study the processing technology of tea and coffee
3. To impart knowledge about the processing of cocoa into various cocoa products

Course Prerequisites:

Topic	Program/CourseName
Basic knowledge of spices	Spices and Condiments

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the status and scope of spice and flavor processing industries in India, including their sources, production, selection criteria, and classification based on origin and physical characteristics.	10	Understanding
CO2	Explain the occurrence, process flow sheet, and mechanisms involved in the manufacturing of coffee powder and instant coffee.	15	Understanding
CO3	Explain the occurrence, process flow, and manufacturing mechanisms of tea, along with the characteristics of different types of tea such as white, green, oolong, and black.	15	Understanding
CO4	Describe the occurrence, processing mechanisms (primary and secondary), and quality control procedures involved in cocoa and chocolate production.	20	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the status and scope of spice and flavor processing industries in India, including their sources, production, selection criteria, and classification based on origin and physical characteristics.		
M1.01	Explain Status and scope of spice and flavor processing industries in India	4	Understanding
M1.02	Discuss sources, production, selection criteria of Spices	3	Understanding
M1.03	Classify spices on the basis of origin, physical characteristic	3	Understanding
Contents: Spices-Status and scope of spice and flavour processing industries in India; sources, production, selection criteria, classification on the basis of origin, physical characteristic			
CO2	Explain the occurrence, process flow sheet, and mechanisms involved in the manufacturing of coffee powder and instant coffee.		
M2.01	Summarize the occurrence of coffee.	4	Understanding
M2.02	Discuss Process flow for the manufacture of coffee powder	5	Understanding
M2.03	Discuss Process flow for the manufacture of instant powder	5	Understanding
	Series Test –I	1	
Contents: Coffee: introduction and Occurrence, Process flow sheet and mechanism of the manufacturing process of coffee powder and instant coffee			

CO3	Explain the occurrence, process flow, and manufacturing mechanisms of tea, along with the characteristics of different types of tea such as white, green, oolong, and black.		
M3.01	Discuss about Tea	5	Understanding
M3.02	Explain Process flow for the manufacture of tea	5	Understanding
M3.03	Differentiate types of tea as white, green, oolong, and black.	5	Understanding
Contents: Tea: introduction and Occurrence, Process flow sheet and mechanism of the manufacturing process of tea, different types of tea- white, green, oolong, black			
CO4	Describe the occurrence, processing mechanisms (primary and secondary), and quality control procedures involved in cocoa and chocolate production.		
M4.01	Summarize introduction and Occurrence of cocoa	6	Understanding
M4.02	Discuss about Process flow for cocoa processing into various products	7	Understanding
M4.03	Explain quality control of chocolates	6	Understanding
	Series Test–II	1	
Contents: Cocoa: introduction and Occurrence, Process flow sheet and mechanism of cocoa processing – primary and secondary processing, Quality control of chocolates			

Text / Reference

T/R	Book Title/Author
R1	Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & Hosahalli S. Ramaswamy: “Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices”, CRC Press, 2003.
R2	K.V. Peter, “Handbook of herbs & spices”, Volume1, ISBN: 0857095684, 9780857095688, Elsevier, 2012.
R3	N. Kumar, “Introduction to Spices, Plantation Crops, Medicinal and Aromatic Plants”, Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018.
R4	Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. “Tea and Tea Products: Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology”, CRC Press, 2008.
R5	Susheela Raghavan, “Handbook Seasoning, Spices & Flavoring”, Second Edition, CRC Press, Publication, 2006.

Online Resources

Sl.No	Website Link
1	http://www.teabreeze.com/Manufacturingprocess.aspx
2	https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices
3	https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/
4	https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-addictive-plants/coca